

17. Rectifiability and densities

In this chapter we shall consider the characterization of rectifiability in terms of densities. We shall prove that condition (1) of Theorem 16.2 is also sufficient for an $\mathcal{H}^m$ measurable set E with $\mathcal{H}^m(E) < \infty$ to be rectifiable. Then we shall discuss the sufficiency of the weaker condition that the density $\Theta^m(E, x)$ exists for $\mathcal{H}^m$ almost all $x \in E$. We shall only prove this for $m = 1$ and sketch a proof for $m = 2$, $n = 3$. The case $m = 1$ is simpler and can also be found in Falconer [4, §3.3] with a different argument. The proof for general dimensions $m \geq 2$ due to Preiss [4] is very complicated and we shall only discuss some ideas behind it. For both results we are going to use tangent measures.

Structure of m -uniform measures

In particular for the latter characterization it is more natural to work with general Radon measures rather than Hausdorff measures on sets. Thus we shall study Radon measures μ for which the positive and finite density

$$(17.1) \quad 0 < \Theta^m(\mu, x) = \lim_{r \downarrow 0} (2r)^{-m} \mu(B(x, r)) < \infty$$

exists μ almost everywhere. The goal is to show that such a μ is rectifiable. In view of Theorem 16.7 one only needs to show that at μ almost all points every tangent measure ν is m -flat. We know from Corollary 14.9 that every such ν is m -uniform so Preiss's theorem would be proven if we could show that every m -uniform measure on $\mathbf{R}^n$ is m -flat. Unfortunately (or fortunately, as things get more exciting) this is not true for general m , see Preiss [4, 3.20], and that is one of the main reasons for the extra work needed. But it is true for $m = 0, 1$ and 2 . We shall prove this for $m = 1$ and sketch a proof in the case $m = 2$, $n = 3$.

The proof that $\Theta^m(E, x) = 1$ for $\mathcal{H}^m$ almost all $x \in E$ implies the rectifiability of E is simpler because of the following fact. It follows from Lemma 14.7 (1) and (2) and Corollary 6.7 that in this case at $\mathcal{H}^m$ almost all points $a \in E$ every $\nu \in \text{Tan}(\mathcal{H}^m \llcorner E, a)$ in addition to being m -uniform, i.e. $\nu(B(x, r)) = cr^m$ for $x \in \text{spt } \nu$, also satisfies

$$\nu(B(x, r)) \leq cr^m \quad \text{for all } x \in \mathbf{R}^n$$

with the same constant c . It turns out that this additional condition forces ν to be m -flat, see Theorem 17.3.

For general n we shall show that the supports of the m -uniform measures are contained in certain conics and that the $(n-1)$ -uniform measures are constant multiples of $\mathcal{H}^{n-1}$ restricted to such conics and we indicate how this information can be used.

In this chapter m and n will be integers with $0 < m < n$.

17.2. Definition. Let $Q: \mathbf{R}^n \rightarrow \mathbf{R}$ be a quadratic polynomial of the form

$$Qx = \sum_{i,j=1}^n a_{ij}x_i x_j \quad \text{for } x = \sum_{i=1}^n x_i e_i$$

where $\{e_1, \dots, e_n\}$ is an orthonormal basis of $\mathbf{R}^n$. The trace of Q , which is independent of the basis $\{e_1, \dots, e_n\}$, is

$$\text{Tr } Q = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n Qe_i.$$

17.3. Theorem. Let ν be an m -uniform measure on $\mathbf{R}^n$ such that $0 \in \text{spt } \nu$ and

$$(1) \quad \nu(B(x, r)) = (2r)^m \quad \text{for } x \in \text{spt } \nu, \ 0 < r < \infty.$$

Then there is a polynomial $P: \mathbf{R}^n \rightarrow \mathbf{R}$ such that $P \not\equiv 0$, the degree of P is 2 and

$$\text{spt } \nu \subset \{x : P(x) = 0\}.$$

If, in addition,

$$(2) \quad \nu(B(x, r)) \leq (2r)^m \quad \text{for } x \in \mathbf{R}^n, \ 0 < r < \infty,$$

then there is $V \in G(n, m)$ such that

$$\nu = \mathcal{H}^m \llcorner V.$$

Proof. For $r > 0$ define $b_r \in \mathbf{R}^n$ by (recall Exercise 1.7)

$$b_r \cdot x = \frac{m+2}{2^{m+1}r^{m+2}} \int_{B(r)} (r^2 - |y|^2)(x \cdot y) \, d\nu y \quad \text{for } x \in \mathbf{R}^n,$$

and a quadratic polynomial Q_r by

$$Q_r x = \frac{m+2}{2^m r^{m+2}} \int_{B(r)} (x \cdot y)^2 \, d\nu y.$$

Since $0 \in \text{spt } \nu$, (1) implies

$$(3) \quad |Q_r x| \leq (m+2)|x|^2 \quad \text{for } x \in \mathbf{R}^n.$$

Hence it follows that there is a sequence $r_i \rightarrow \infty$ for which Q_{r_i} converges to a quadratic polynomial Q ; to achieve this we only need the convergence of, for example, $Q_{r_i}(e_j + e_k)$, $j, k = 1, \dots, n$, for the basis vectors e_j . By Theorem 1.15 and (1) we can compute

$$\begin{aligned} \int_{B(r)} |y|^2 d\nu y &= \int_0^{r^2} \nu(\{y : t \leq |y|^2 \leq r^2\}) dt \\ &= \int_0^{r^2} 2^m(r^m - t^{m/2}) dt = \frac{m}{m+2} 2^m r^{m+2}. \end{aligned}$$

This gives

$$\text{Tr } Q = \lim_{i \rightarrow \infty} \text{Tr } Q_{r_i} = \lim_{i \rightarrow \infty} \frac{m+2}{2^m r_i^{m+2}} \int_{B(r_i)} |y|^2 d\nu y = m.$$

Let $x \in \mathbf{R}^n$ and $r > 2|x|$. We consider the balls

$$B_1 = B(x, r - |x|), \quad B_2 = B(r), \quad B_3 = B(x, r), \quad B_4 = B(x, r + |x|),$$

and put

$$J_i = \int_{B_i} (r^2 - |y - x|^2)^2 d\nu y \quad \text{for } i = 1, \dots, 4.$$

Then we have $B_1 \subset B_2 \subset B_4$ and $B_1 \subset B_3 \subset B_4$, whence $J_1 \leq J_2 \leq J_4$ and $J_1 \leq J_3 \leq J_4$. Thus

$$(4) \quad |J_2 - J_3| \leq J_4 - J_1.$$

For $y \in B_4 \setminus B_1$ we have

$$r - |x| \leq |y - x| \leq r + |x|,$$

which implies

$$-|x|^2 - 2r|x| \leq r^2 - |y - x|^2 \leq 2r|x| - |x|^2,$$

and because $|x| \leq r$, we get

$$|r^2 - |y - x|^2| \leq 3r|x|.$$

This gives

$$\begin{aligned}
 J_4 - J_1 &= \int_{B_4 \setminus B_1} (r^2 - |y - x|^2)^2 d\nu y \leq 9r^2 |x|^2 \nu(B_4 \setminus B_1) \\
 &\leq 9r^2 |x|^2 (\nu(B(r + 2|x|)) \setminus \nu(B(r - 2|x|))) \\
 &= 9 \cdot 2^{m+1} r^{m+1} |x|^3 \frac{(1 + 2|x|/r)^m - (1 - 2|x|/r)^m}{2|x|/r} \\
 &\leq 9 \cdot 2^{m+1} 2^m m r^{m+1} |x|^3,
 \end{aligned}$$

by the elementary inequality $(1 + t)^m - (1 - t)^m \leq m 2^m t$ for $0 < t < 1$. Thus by (4),

$$(5) \quad |J_2 - J_3| \leq 18m 4^m r^{m+1} |x|^3.$$

Computing by Theorem 1.15 and using (1), we find for $x \in \text{spt } \nu$ with $2|x| < r$,

$$\begin{aligned}
 J_3 &= \int_{B(x, r)} (r^2 - |y - x|^2)^2 d\nu y \\
 &= \int_0^{r^4} \nu(\{y : t \leq (r^2 - |y - x|^2)^2\}) dt \\
 &= \int_0^{r^4} \nu\left(B\left(x, \sqrt{r^2 - \sqrt{t}}\right)\right) dt = 2^m \int_0^{r^4} (r^2 - \sqrt{t})^{m/2} dt \\
 &= 2^m \int_0^{r^4} \nu\left(B\left(\sqrt{r^2 - \sqrt{t}}\right)\right) dt = \int_{B(r)} (r^2 - |y|^2)^2 d\nu y.
 \end{aligned}$$

Note that if (2) holds we have this with inequality for all $x \in \mathbf{R}^n$ with $2|x| < r$:

$$J_3 \leq \int_{B(r)} (r^2 - |y|^2)^2 d\nu y.$$

Using the equation $|y - x|^2 = |y|^2 + |x|^2 - 2x \cdot y$ we then obtain for $x \in U(r/2)$

$$\begin{aligned}
 (6) \quad J_2 - J_3 &\geq \int_{B(r)} ((r^2 - |y - x|^2)^2 - (r^2 - |y|^2)^2) d\nu y \\
 &= 4 \int_{B(r)} (r^2 - |y|^2)(y \cdot x) d\nu y + 4 \int_{B(r)} (y \cdot x)^2 d\nu y \\
 &\quad - 2|x|^2 \int_{B(r)} (r^2 - |y|^2) d\nu y + |x|^4 \nu(B(r)) \\
 &\quad - 4|x|^2 \int_{B(r)} (y \cdot x) d\nu y.
 \end{aligned}$$

For $x \in \text{spt } \nu$ (without assuming (2)), this holds as equality. Clearly,

$$(7) \quad |x|^4 \nu(B(r)) = 2^m r^m |x|^4 \leq 2^m r^{m+1} |x|^3,$$

$$(8) \quad \left| 4|x|^2 \int_{B(r)} y \cdot x \, d\nu y \right| \leq 2^{m+2} r^{m+1} |x|^3.$$

Let $x \in \text{spt } \nu$. Applying (5), (7), (8) and the equality in (6), we find that there is a constant C such that

$$\begin{aligned} & \left| 4 \int_{B(r)} (r^2 - |y|^2)(y \cdot x) \, d\nu y + 4 \int_{B(r)} (y \cdot x)^2 \, d\nu y - 2|x|^2 \int_{B(r)} (r^2 - |y|^2) \, d\nu y \right| \\ & \leq \frac{2^{m+2}}{m+2} C r^{m+1} |x|^3. \end{aligned}$$

Dividing by

$$\frac{2^{m+2}}{m+2} r^{m+2} = 2 \int_{B(r)} (r^2 - |y|^2) \, d\nu y$$

we obtain

$$(9) \quad |2b_r \cdot x + Q_r x - |x|^2| \leq C|x|^3/r$$

for $x \in \text{spt } \nu$ and $r > 2|x|$. This gives

$$\lim_{i \rightarrow \infty} b_{r_i} \cdot x = \frac{1}{2}(|x|^2 - Q(x)) \quad \text{for } x \in \text{spt } \nu.$$

From the definition of b_r we see easily that it belongs to the linear span of $\text{spt } \nu$. Hence it follows that b_{r_i} converges to some $b \in \mathbf{R}^n$ and (9) yields

$$\text{spt } \nu \subset K = \{x \in \mathbf{R}^n : Qx - |x|^2 + 2b \cdot x = 0\}.$$

Denoting $Q_1 x = Qx - |x|^2$ we have $\text{Tr } Q_1 = m - n$ so that $Q_1 \neq 0$. Letting $Px = Q_1 x + 2b \cdot x$, the first part of the theorem follows.

Suppose now that (2) holds. As above we infer from (6) and (5) that

$$(10) \quad 2b_r \cdot x + Q_r x - |x|^2 \leq C|x|^3/r$$

for $x \in \mathbf{R}^n$ with $2|x| < r$. This gives again that $b_{r_i} \rightarrow b$ which in view of (3) satisfies

$$2b \cdot x \leq (m+3)|x|^2 \quad \text{for all } x \in \mathbf{R}^n.$$

Clearly this implies $b = 0$, whence by (10), $Qx \leq |x|^2$ for $x \in \mathbf{R}^n$. Since $\text{Tr } Q = m$, it follows that

$$K = \{x : Qx = |x|^2\}$$

must be an m -plane. This can be seen by taking an orthonormal basis $\{e_1, \dots, e_n\}$ of $\mathbf{R}^n$ such that $Qx = \sum_{i=1}^n a_i x_i^2$ for $x = \sum_{i=1}^n x_i e_i$. Then $0 \leq a_i \leq 1$ and $\sum_{i=1}^n a_i = m$. Obviously $Qx < |x|^2$ for $x \in \mathbf{R}^n \setminus V$ where V is the linear span of $\{e_i : a_i = 1\}$. Thus $\text{spt } \nu \subset V$ and so, as ν is m -uniform, $\dim V \geq m$. But $\text{Tr } Q = m$ implies then $\dim V = m$ and $a_i = 0$ when $e_i \notin V$. Thus $Qx = \sum_{e_i \in V} x_i^2$ and $K = V$.

Finally Theorem 14.11 together with (1) implies $\nu = \mathcal{H}^m \llcorner V$. $\square$

17.4. Remark. The above proof of Theorem 17.3 is essentially from Kowalski and Preiss [1]. In codimension 1 one can use Theorem 17.3 to get a great deal more information about uniform measures.

Let ν be an $(n-1)$ -uniform measure on $\mathbf{R}^n$ with

$$\nu(B(x, r)) = (2r)^{n-1} \quad \text{for } x \in \text{spt } \nu, \ 0 < r < \infty.$$

Then there is a non-zero quadratic polynomial $P: \mathbf{R}^n \rightarrow \mathbf{R}$ such that setting $K = \{x : P(x) = 0\}$,

$$\nu = \mathcal{H}^{n-1} \llcorner K.$$

To see this observe first that by Exercise 1, $\nu = \mathcal{H}^{n-1} \llcorner \text{spt } \nu$. So one is left to show that $\text{spt } \nu = K = \{x : P(x) = 0\}$. For this, see Kowalski and Preiss [1]. In fact, Kowalski and Preiss prove much more about K . They show that either K is a hyperplane, or $n \geq 4$ and, after a rotation and translation,

$$K = \{x \in \mathbf{R}^n : x_4^2 = x_1^2 + x_2^2 + x_3^2\}.$$

Conversely, a direct computation shows that for such a K , $\mathcal{H}^{n-1} \llcorner K$ is $(n-1)$ -uniform.

In general codimensions the structure of the supports of m -uniform measures is quite open. Recall however the remarks following Theorem 3.4 on the more general uniformly distributed measures.

Next we shall prove that every 1-uniform measure on $\mathbf{R}^n$ is 1-flat. It is possible to give an elementary proof for this, but we shall continue to develop the proof of Theorem 17.3. We shall also sketch a differential geometric argument showing that 2-uniform measures on $\mathbf{R}^3$ are 2-flat. For the proof that 2-uniform measures on any $\mathbf{R}^n$ are 2-flat, see Preiss [4, 3.17]; we shall explain some ideas behind this at the end of this chapter.

17.5. Theorem. Let $m = 1$ or $m = 2$ and let ν be an m -uniform measure on $\mathbf{R}^n$. Then ν is m -flat.

Proof for $m = 1$. Let ν be a 1-uniform measure on $\mathbf{R}^n$ normalized so that $0 \in \text{spt } \nu$ and

$$\nu(B(x, r)) = 2r \quad \text{for } x \in \text{spt } \nu, \quad 0 < r < \infty.$$

We return to the set-up of the proof of Theorem 17.3. We had there

$$\text{spt } \nu \subset \{x \in \mathbf{R}^n : 2b \cdot x = |x|^2 - Qx\},$$

where $b \in \mathbf{R}^n$, and we can again express Q as

$$Qx = \sum_{i=1}^n a_i x_i^2 \quad \text{for } x = \sum_{i=1}^n x_i e_i$$

with some orthonormal basis $\{e_1, \dots, e_n\}$ of $\mathbf{R}^n$ choosing the order so that $a_1 \geq a_2 \geq \dots \geq a_n \geq 0$. Moreover we had $\text{Tr } Q = \sum_{i=1}^n a_i = 1$ and for some sequence $r_j \rightarrow \infty$,

$$(1) \quad b \cdot x = \lim_{j \rightarrow \infty} \frac{3}{4} r_j^{-3} \int_{B(r_j)} (r_j^2 - |y|^2)(x \cdot y) d\nu y \quad \text{for } x \in \mathbf{R}^n.$$

We claim that $a_1 = 1$ and $a_i = 0$ for $i = 2, \dots, n$, whence $Qx = x_1^2$. If $a_1 < 1$, we would have for $x \in \text{spt } \nu$,

$$2b \cdot x = |x|^2 - Qx \geq |x|^2 - a_1 |x|^2,$$

whence $|x| \leq 2|b|/(1 - a_1)$. Thus $\text{spt } \nu$ would be bounded, which is impossible by the 1-uniformity of ν . Hence $a_1 \geq 1$ and then $\sum_{i=1}^n a_i = 1$ implies $a_1 = 1$ and $a_i = 0$ for $i = 2, \dots, n$. Thus we have

$$(2) \quad \text{spt } \nu \subset \{x : 2b \cdot x = x_2^2 + \dots + x_n^2\}.$$

Next we show that $b_1 = b \cdot e_1 = 0$. To see this consider the 1-uniform measures $\mu_r = r^{-1} T_{0, r\sharp} \nu$, $r > 0$, which satisfy $0 \in \text{spt } \mu_r$, $\mu_r(B(x, \varrho)) = 2\varrho$ for $x \in \text{spt } \mu_r$, $0 < \varrho < \infty$, and

$$\begin{aligned} \text{spt } \mu_r &\subset \{x : 2b \cdot (rx) = r^2(x_2^2 + \dots + x_n^2)\} \\ &= \{x : (2b \cdot x)/r = x_2^2 + \dots + x_n^2\} \subset \{x : b \cdot x \geq 0\}. \end{aligned}$$

By Theorem 1.23 there is a sequence $r'_j \rightarrow \infty$ such that $(\mu_{r'_j})$ converges to a Radon measure μ which is readily seen to be 1-uniform with $\mu(B(x, \varrho)) = 2\varrho$ for $x \in \text{spt } \mu$ and $0 < \varrho < \infty$, and with

$$\begin{aligned}\text{spt } \mu &\subset \{x : x_2 = \cdots = x_n = 0\} \cap \{x : x \cdot b \geq 0\} \\ &= \{x_1 e_1 : x_1 b_1 \geq 0\}.\end{aligned}$$

Since the support of μ is contained in the line $L = \{te_1 : t \in \mathbf{R}\}$, $\mu = \mathcal{H}^1 \llcorner L$ by Theorem 14.11. In particular $\text{spt } \mu = L$ and so $b_1 = 0$.

(The above argument shows that if $r'_j \rightarrow \infty$ and $(r'_j)^{-1} T_{0, r'_j \sharp} \nu$ converges weakly, then the limit is $\mathcal{H}^1 \llcorner L$. In fact, this implies that $r^{-1} T_{0, r \sharp} \nu \xrightarrow{w} \mathcal{H}^1 \llcorner L$ as $r \rightarrow \infty$, which in the language of Preiss means that ν is flat at infinity, see Preiss [4, 3.12 (9)].)

We have now shown that the support of ν is contained in a cylinder with a bounded basis:

$$\text{spt } \nu \subset K = \{x : 2(b_2 x_2 + \cdots + b_n x_n) = x_2^2 + \cdots + x_n^2\} = \mathbf{R} \times B.$$

Let $d = d(B)$. Then $d \leq 4|b|$. Let $S_+ = \{x \in \text{spt } \nu : x_1 > 0\}$ and $S_- = \{x \in \text{spt } \nu : x_1 < 0\}$. The fact that μ above equals $\mathcal{H}^1 \llcorner L$ implies that both S_+ and S_- are unbounded. It is easy to verify this also directly.

Let $r_0 > d$ be such that $B(r_0) \cap S_+ \neq \emptyset$ and $B(r_0) \cap S_- \neq \emptyset$. We claim that

$$(3) \quad S(r) \cap S_+ \neq \emptyset \quad \text{and} \quad S(r) \cap S_- \neq \emptyset \quad \text{for } r > r_0.$$

Suppose for example that there are radii $r > r_0$ for which $S(r) \cap S_+ = \emptyset$. Then we can find $r > r_0$ and $\delta > 0$ such that there is $b \in S(r) \cap S_+$ and $\text{spt } \nu \cap B(r + \delta) \setminus B(r) = \emptyset$. Let $e = b/|b|$ and for $\varrho > 0$,

$$C(\varrho) = \{x : \varrho < |b - x| \leq 2\varrho, (b - x) \cdot e \geq |b - x|/4\}.$$

Then

$$(4) \quad \text{spt } \nu \cap B(b, \delta) \subset \{x : (b - x) \cdot e \geq 0\}$$

and for $0 < \varrho < \delta/2$, by a simple application of the triangle inequality,

$$\text{spt } \nu \cap B(b, 2\varrho) \setminus B(b, \varrho) \setminus C(\varrho) \subset B(r) \setminus B(r - \varrho/2),$$

whence

$$(5) \quad \nu(C(\varrho)) \geq \nu(B(b, 2\varrho) \setminus B(b, \varrho)) - \nu(B(r) \setminus B(r - \varrho/2)) = \varrho.$$

Let $\lambda \in \text{Tan}(\nu, b)$. Then (4) and (5) imply

$$(6) \quad \text{spt } \lambda \subset \{x : x \cdot e \leq 0\},$$

$$(7) \quad \text{spt } \lambda \cap (B(2R) \setminus U(R)) \cap \{x : x \cdot e \leq -|x|/4\} \neq \emptyset$$

for $0 < R < \infty$. Since λ is 1-uniform, there are, by what we have proved so far about 1-uniform measures, $u \in S^{n-1}$ and a bounded set $C \subset \{x : x \cdot u = 0\}$ such that $\text{spt } \lambda \subset \{tu + v : t \in \mathbf{R}, v \in C\}$ and that both $\{tu + v \in \text{spt } \lambda : t > 0\}$ and $\{tu + v \in \text{spt } \lambda : t < 0\}$ are unbounded. Clearly this contradicts (6) and (7). Thus we have verified (3).

Next we show that

$$(8) \quad \text{spt } \nu \setminus B(r_0) \subset \{te_1 : t \in \mathbf{R}\} = L.$$

Suppose this is not true and, for example, that $S_+ \setminus B(r_0)$ is not contained in L . Using (3) and the fact that $d < r_0$, we can then find points $x, y \in S_+$ such that $2r_0 < |x| < |y|$, $|y - x| < r_0$, and that $0, x$ and y do not lie on the same line. Letting $\varepsilon = (|y| - |x|)/2$ and $\delta = |y - x|/2$, we have then $\varepsilon < \delta$. Moreover,

$$\begin{aligned} B(x, \delta) \cup B(y, \delta) &\subset B(|y| + \delta) \setminus U(|x| - \delta) \\ &= B(r + \varepsilon + \delta) \setminus U(r - \varepsilon - \delta), \end{aligned}$$

where $r = |x| + \varepsilon$. By (3) we find $z \in S(r) \cap S_-$. Then

$$(9) \quad B(x, \delta) \cup B(y, \delta) \cup B(z, \varepsilon + \delta) \subset B(r + \varepsilon + \delta) \setminus U(r - \varepsilon - \delta),$$

and the balls $U(x, \delta)$, $U(y, \delta)$ and $U(z, \varepsilon + \delta)$ are disjoint (the first two are contained in $\{x : x_1 > 0\}$ and the third one in $\{x : x_1 < 0\}$). Taking the ν measure of the sets in (9), we obtain $6\delta + 2\varepsilon \leq 4\delta + 4\varepsilon$, whence $\delta \leq \varepsilon$, which is a contradiction. Thus (8) holds.

Since $b_1 = 0$, (8) tells us that $b \cdot y = 0$ for $y \in \text{spt } \nu \setminus B(r_0)$. Therefore we see from (1) that $b = 0$. Consequently (2) yields that $\text{spt } \nu \subset L$, whence $\nu = \mathcal{H}^1 \llcorner L$ by Theorem 14.11. $\square$

Sketch of proof for $m = 2 = n - 1$: From Theorem 17.3 we know that $\text{spt } \nu$ is contained in some quadratic conic $K = \{x : P(x) = 0\}$ such that

$K \neq \mathbf{R}^3$. Outside a closed singular set S of σ -finite $\mathcal{H}^1$ measure K is locally a smooth surface. By Kowalski and Preiss [1, Theorem 2.2 and (2.19)], for any $x \in K \setminus S$ and sufficiently small $r > 0$,

$$\mathcal{H}^2(K \cap B(x, r)) = 4r^2(1 + \frac{1}{32}(\lambda_1 - \lambda_2)^2) + O(r^4)$$

where λ_1 and λ_2 are the principal curvatures of K at x . (For balls in the intrinsic metric of the smooth surface $K \setminus S$ an analogous formula is classical in differential geometry.) Referring to Exercise 1 we have on the other hand,

$$\mathcal{H}^2(\text{spt } \nu \cap B(x, r)) = 4r^2$$

for $x \in \text{spt } \nu$, $0 < r < \infty$. Simple arguments like those used in the proof of Lemma 14.7 show that $K \cap B(x, r) = \text{spt } \nu \cap B(x, r)$ for $x \in \text{spt } \nu \setminus S$ and small r and so $\lambda_1 = \lambda_2$. By a classical result of differential geometry this means that $K \setminus S$ is locally contained either in a plane or a sphere. Since K is a zero-set of a quadratic polynomial, it follows that $\text{spt } \nu$ is contained in a finite union of planes and spheres. After this it is an easy exercise to show that $\text{spt } \nu$ is contained in a plane V , whence $\nu = \mathcal{H}^2 \llcorner V$ by Theorem 14.11. $\square$

Rectifiability and density one

We shall now use Theorem 17.3 to deduce a density characterization of m -rectifiable sets.

17.6. Theorem. *Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$.*

- (1) *E is m -rectifiable if and only if the density $\Theta^m(E, x)$ exists and equals 1 for $\mathcal{H}^m$ almost all $x \in E$.*
- (2) *E is purely m -unrectifiable if and only if $\Theta_*^m(E, x) < 1$ for $\mathcal{H}^m$ almost all $x \in E$.*

Proof. Due to Theorem 6.2 these statements are equivalent. We prove (1). From Theorem 16.2 we already know that $\Theta^m(E, x) = 1$ almost everywhere on E if E is m -rectifiable. To prove the converse we combine Theorems 16.5 and 17.3. First applying Corollary 6.7, Lemma 14.7 and Corollary 14.9 to $\mathcal{H}^m \llcorner E$, we see that for $\mathcal{H}^m$ almost all $a \in E$ every $\nu \in \text{Tan}(\mathcal{H}^m \llcorner E, a)$ after a normalization satisfies conditions (1) and (2) of Theorem 17.3, whence $\nu = \mathcal{H}^m \llcorner V$ for some $V \in G(n, m)$. Consequently E is m -rectifiable by Theorem 16.5. $\square$

17.7. Remarks. Using Theorem 6.6 one can check that the density $\Theta^m(E, x)$ with respect to the Hausdorff measure could be replaced in Theorem 17.6 by the density $\Theta^m(\mathcal{S}^m \llcorner E, x)$ with respect to the spherical measure.

Chlebík [1] improved Theorem 17.6 by proving that there is a constant $c(m) < 1$, depending only on m and not on n , such that if $\Theta_*^m(E, x) \geq c(m)$ for $\mathcal{H}^m$ almost all $x \in E$, then E is m -rectifiable. Chlebík's proof, which was based on the proofs in Marstrand [3] and Mattila [1], works also for subsets E of some infinite dimensional Hilbert spaces.

For general m very little is known about the best possible constant $c(m)$ as above. But for $m = 1$ there is much more information. For subsets of $\mathbf{R}^2$ Besicovitch showed that

$$(1) \quad \Theta_*^1(E, x) \geq 3/4 \quad \text{for } \mathcal{H}^1 \text{ almost all } x \in E$$

implies the 1-rectifiability of E , see Besicovitch [4] and Falconer [4, 3.24]. Preiss and Tišer [3] improved this by showing that $3/4$ can be replaced by $(2 + \sqrt{46})/12 \approx 0.732$. Their proof works for subsets of general metric spaces. Besicovitch [1] gave an example of a purely 1-unrectifiable subset of $\mathbf{R}^2$ for which $\Theta_*^1(E, x) = 1/2$ for $\mathcal{H}^1$ almost all $x \in E$ and conjectured that $3/4$ could be replaced by $1/2$ in (1). This conjecture is still open.

Preiss's theorem

Besicovitch [4] also proved that for $\mathcal{H}^1$ measurable sets $E \subset \mathbf{R}^2$ with $\mathcal{H}^1(E) < \infty$ the rectifiability of E follows from the almost everywhere existence of the density $\Theta^1(E, x)$. This was extended for general one-dimensional Borel measures in $\mathbf{R}^2$ by Morse and Randolph [1] and in $\mathbf{R}^n$ by Moore [1]. The corresponding question for m -dimensional sets and measures remained open for a long time until Preiss [4] solved it completely. We shall now state his theorem and prove it in the case $m = 1$ or 2 (modulo the gaps in the proof of Theorem 17.5). Recall the definition of a rectifiable measure from 16.6.

17.8. Theorem. *Let μ be a Radon measure on $\mathbf{R}^n$ such that the density $\Theta^m(\mu, x)$ exists and is positive and finite for μ almost all $x \in \mathbf{R}^n$. Then μ is m -rectifiable.*

In fact, somewhat less suffices: there is a constant $c(n, m) > 1$ such that if

$$0 < \Theta^{*m}(\mu, x) \leq c(n, m) \Theta_*^m(\mu, x) < \infty$$

for μ almost all $x \in \mathbf{R}^n$, then μ is m -rectifiable, see Preiss [4, 5.7]. Combining Theorem 17.8 with Theorem 16.2 we have immediately for sets

17.9. Corollary. Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. Then E is m -rectifiable if and only if the density $\Theta^m(E, x)$ exists for $\mathcal{H}^m$ almost all $x \in E$.

For $m = 1$ and $m = 2$ Theorem 17.8 follows now easily. By Corollary 14.9 at μ almost all points all tangent measures of μ are m -uniform, whence by Theorem 17.5 they are m -flat. The rectifiability of μ follows then from Theorem 16.7.

17.10. A sketch of the proof of Preiss's theorem. We shall now explain some basic ideas which Preiss used to prove Theorem 17.8. If μ is as in Theorem 17.8 we know by Corollary 14.9 that at μ almost all points the tangent measures of μ are m -uniform and we should know in view of Theorem 16.7 that they are m -flat. As noted before, for $m = 1$ and $m = 2$ this follows from the fact that m -uniform measures are m -flat but for $m \geq 3$ we need some other information about m -uniform measures. Briefly this additional information says that if an m -uniform measure is not m -flat, then it has to be rather far away from all flat measures. We know from Theorem 14.18 that μ has flat tangent measures at μ almost all points a . Hence it is sufficient to show that $\text{Tan}(\mu, a)$ is in a suitable sense connected to guarantee that μ has only flat tangent measures μ almost everywhere. Such connectedness is not very difficult to prove; see Preiss [4, Theorem 2.6] and also Exercise 3.

We give now some details. Let ν be an m -uniform measure on $\mathbf{R}^n$ normalized so that $0 \in \text{spt } \nu$ and

$$\nu(B(x, r)) = (2r)^m \quad \text{for } x \in \text{spt } \nu \text{ and } 0 < r < \infty.$$

In fact, much of what will be said below holds for more general uniformly distributed measures, see Preiss [4, Section 3].

Instead of the integrals over balls such as $\int_{B(r)} (y \cdot z)^k d\nu z$, which we used before, it is better to investigate integrals $\int (y \cdot z)^k e^{-s|z|^2} d\nu z$, $s > 0$. Note that since $e^{-s|z|^2}$ is small when $s|z|^2$ is large, the latter integral can be considered as a smoothing of the former for $s = r^{-2}$. In particular,

$$I(s) = \int e^{-s|z|^2} d\nu z = cs^{-m/2} = c2^{-m}\nu(B(s^{-1/2})).$$

Define for $k = 1, 2, \dots$

$$b_{k,s}(u) = (2s)^k (I(s) k!)^{-1} \int (z \cdot u)^k e^{-s|z|^2} d\nu z, \quad u \in \mathbf{R}^n.$$

The behaviour of $b_{k,s}$ as $s \downarrow 0$ will be most important. An easy estimate is

$$(1) \quad |b_{k,s}(u)| \leq 2^k 5^n k^{k/2} (k!)^{-1} s^{k/2} |u|^k$$

which is based only on the inequality $\nu(B(r)) \leq 2^m r^m$. A more delicate and crucial piece of information is that for odd k this can be improved to

$$(2) \quad |b_{k,s}(u)| \leq c_k s^{(k+1)/2} |u|^k \quad \text{for odd } k.$$

(Note that if ν were flat, $b_{k,s}$ would be zero for odd k .) Moreover, $b_{k,s}$ has a useful Taylor expansion. To see this one first verifies that

$$(3) \quad \sum_{j=1}^{\infty} (s^j/j!) \int e^{-s|z|^2} (2^j(z \cdot x)^j - |x|^{2j}) d\nu z = 0 \quad \text{for } x \in \text{spt } \nu, s > 0.$$

This is obtained by observing that

$$\int e^{-s|z-x|^2} d\nu = \int e^{-s|z|^2} d\nu z \quad \text{for } x \in \text{spt } \nu,$$

whence

$$\begin{aligned} & \int \sum_{j=1}^{\infty} (s^j/j!) e^{-s|z|^2} (2^j(z \cdot x)^j - |x|^{2j}) d\nu z \\ &= \int (e^{-s|z|^2+2sz \cdot x} - e^{-s|z|^2+s|x|^2}) d\nu z \\ &= e^{s|x|^2} \int (e^{-s|z-x|^2} - e^{-s|z|^2}) d\nu z = 0. \end{aligned}$$

Thus one obtains (3) by interchanging integration and summation, which can easily be justified. The equation (3) leads for $q = 1, 2, \dots$ to

$$(4) \quad \left| \sum_{k=1}^{2q} b_{k,s}(x) - \sum_{k=1}^q \frac{s^k |x|^{2k}}{k!} \right| \leq 5^{n+9} (s|x|^2)^{q+1/2} \quad \text{for } x \in \text{spt } \nu.$$

For $s|x|^2 > 1$ this is crude and based only on (1) which yields even that

$$\sum_{k=1}^{2q} |b_{k,s}(x)| + \sum_{k=1}^q \frac{s^k |x|^{2k}}{k!} \leq 5^{n+9} (s|x|^2)^{q+1/2}.$$

For $s|x|^2 \leq 1$ one uses (3) to get

$$\left| \sum_{k=1}^{2q} b_{k,s}(x) - \sum_{k=1}^q \frac{s^k |x|^{2k}}{k!} \right| \leq \sum_{k=2q+1}^{\infty} |b_{k,s}(x)| + \sum_{k=q+1}^{\infty} \frac{s^k |x|^{2k}}{k!},$$

which can be estimated by means of (1). From (4) one obtains in particular (2). For example when $k = 1$ we infer with $q = 1$ in (4),

$$\begin{aligned} |b_{1,s}(x)| &\leq b_{2,s}(x) + s|x|^2 + 5^{n+9}(s|x|^2)^{3/2} \\ &\leq 4 \cdot 5^n s|x|^2 + s|x|^2 + 5^{n+9}(s|x|^2)^{3/2} \end{aligned}$$

by the easier estimate (1) for $b_{2,s}$.

Using (4) it is possible to get the expansion, see Preiss [4, Theorem 3.6],

$$(5) \quad b_{k,s} = \sum_{j=1}^q \frac{s^j b_k^{(j)}}{j!} + o(s^q) \quad \text{as } s \downarrow 0$$

where $b_k^{(j)} = b_k^{(j)}(\nu)$ is a polynomial of degree k , $b_k^{(j)} = 0$ when $k > 2j$, and

$$(6) \quad \sum_{k=1}^{2q} b_k^{(q)}(x) = |x|^{2q} \quad \text{for } x \in \text{spt } \nu, \quad q = 1, 2, \dots$$

Thus for example

$$(7) \quad b_1^{(1)}(u) = \lim_{s \downarrow 0} s^{-1} b_{1,s}(u) = \lim_{s \downarrow 0} 2I(s)^{-1} \int (z \cdot u) e^{-s|z|^2} d\nu z,$$

which corresponds to $u \mapsto b \cdot u$ in the proof of Theorem 17.3, and

$$(8) \quad b_2^{(1)}(u) = \lim_{s \downarrow 0} s^{-1} b_{2,s}(u) = \lim_{s \downarrow 0} 2s I(s)^{-1} \int (z \cdot u)^2 e^{-s|z|^2} d\nu z,$$

which corresponds to Q in the proof of Theorem 17.3.

The first application of the expansions of $b_{k,s}$ is to show that ν has a unique tangent measure at infinity; see Preiss [4, Theorem 3.11]. This means that there is a Radon measure λ such that for every $x \in \mathbf{R}^n$

$$(9) \quad \lim_{r \rightarrow \infty} r^{-m} T_{x, r\sharp} \nu = \lambda.$$

That λ is independent of x is rather obvious; the essential part is the convergence at some point, say at 0. This can be reduced to the convergence of the measures

$$\nu_s : A \mapsto \int_A e^{-s|z|^2} d\nu z / I(s)$$

as $s \downarrow 0$. To see that the measures ν_s converge one can apply (5) to find that

$$\lim_{s \downarrow 0} \int (z \cdot u)^k d\nu_s = \begin{cases} 1 & \text{for } k = 0, \\ 0 & \text{for } k \text{ odd,} \\ 2^{-k} k! b_k^{(k/2)}(u) & \text{for } k \geq 2 \text{ even.} \end{cases}$$

But the convergence of the moments implies the convergence; this can be verified with the help of the Fourier transform.

For more details on the following, see Preiss [4, Theorem 3.10]. Let λ be given by (9). Then λ is m -uniform. It is easy to check the scaling property

$$(10) \quad T_{0,r\sharp} \lambda = r^m \lambda \quad \text{for } r > 0.$$

This gives $b_{s,k}(\lambda) = s^{k/2} b_{k,1}(\lambda)$ for $s > 0$, $k = 1, 2, \dots$. In view of (5) we have then

$$(11) \quad b_{2k-1,s}(\lambda) = 0 \quad \text{and} \quad b_{2k,s}(\lambda) = s^k b_{2k}^{(k)}(\lambda) / k!$$

for $k = 1, 2, \dots$. Recalling also (6) we see that

$$(12) \quad \text{spt } \lambda \subset \{x : b_{2k}^{(k)}(\lambda)(x) = |x|^{2k}\}.$$

With this information one can compute for $x \in \text{spt } \lambda$, $p = 1, 2, \dots$, the integrals $\int (z \cdot x)^p e^{-s|z|^2} d\nu z$ and hence also the integrals $\int f(z \cdot x) d\nu z$, when f is a non-negative Borel function on $\mathbf{R}$. These integrals agree with the corresponding integrals for m -flat measures. This gives in particular for $m \geq 2$ that

$$(13) \quad \lambda(\{z : |z \cdot x| \leq 1\}) = \infty \quad \text{for } x \in \text{spt } \lambda.$$

From (8) and (11) one infers, see Preiss [4, 3.12],

$$(14) \quad b_2^{(1)}(\nu)(u) = 2 \int (z \cdot u)^2 e^{-|z|^2} d\lambda z / \int e^{-|z|^2} d\lambda z = b_2^{(1)}(\lambda)(u)$$

for $u \in \mathbf{R}^n$, $k = 1, 2, \dots$. In particular,

$$(15) \quad \text{Tr } b_2^{(1)}(\nu) = 2 \int |z|^2 e^{-|z|^2} d\mathcal{L}^m z \Big/ \int e^{-|z|^2} d\mathcal{L}^m z = m.$$

Let us say that ν is flat at infinity if the tangent measure λ of ν at infinity is m -flat. Before finishing the proof that 1-uniform measures are 1-flat we saw and used the fact that they are flat at infinity. This is also the next step for 2-uniform measures; see Preiss [4, Theorem 3.14]. To prove this we work with $b_2^{(1)} = b_2^{(1)}(\nu)$ as we worked with Q before. Let $a_1 \geq \dots \geq a_n \geq 0$ and an orthonormal basis $\{e_1, \dots, e_n\}$ of $\mathbf{R}^n$ be such that $b_2^{(1)}(x) = \sum_{i=1}^n a_i x_i^2$. Then by (15), $\sum_{i=1}^n a_i = \text{Tr } b_2^{(1)} = m$.

By the scaling property (10) we can choose $y_1 \in S^{n-1} \cap \text{spt } \lambda$. Then by (14) and (12), $a_1 \geq b_2^{(1)}(\nu)(y_1) = b_2^{(1)}(\lambda)(y_1) = 1$. If $m = 1$, we can conclude as before from $\sum_i a_i = 1$ that $a_1 = 1$ and $a_i = 0$ for $i \geq 2$. If $m = 2$, one can use (13) and (10) to show that there exists $y_2 \in S^{n-1} \cap \text{spt } \nu$ such that $y_1 \cdot y_2 = 0$. Thus $a_1 \geq b_2^{(1)}(y_1) = 1$ and $a_2 \geq b_2^{(1)}(y_2) = 1$. Since $\sum_i a_i = 2$, we conclude $a_1 = a_2 = 1$ and $a_i = 0$ for $i \geq 3$. This gives

$$b_2^{(1)}(\lambda)(x) = b_2^{(1)}(\nu)(x) = x_1^2 + x_2^2,$$

whence (12) implies that $\text{spt } \lambda$ is contained in a 2-plane and so λ is 2-flat.

The selection of y_j 's cannot be continued further and m -uniform measures need not be flat at infinity if $m \geq 3$. However, if ν is sufficiently close to a flat measure then it is flat at infinity. This closeness condition can be expressed as

$$(16) \quad \text{Tr } (b_2^{(1)}(\nu) \llcorner W) = 2 \int |P_W z|^2 e^{-|z|^2} d\lambda z \Big/ \int e^{-|z|^2} d\lambda z < \omega(m)$$

for some $W \in G(n, n-m)$ where $\omega(m)$ is a positive constant depending only on m . Under this assumption one can modify the above proof to show that $a_1 = \dots = a_m = 1$ and $a_i = 0$ for $i > m$ which again yields that λ is m -flat.

Suppose now that ν is flat at infinity; $\lambda = \mathcal{H}^m \llcorner V$ for some $V \in G(n, m)$. Then $b_2^{(1)}(u) = b_2^{(1)}(\nu)(u) = |P_V u|^2$ by (14). Using (6) this gives

$$b_1^{(1)}(x) = |x|^2 - b_2^{(1)}(x) = |P_{V^\perp} x|^2 \quad \text{for } x \in \text{spt } \nu.$$

From this one easily concludes that $b_1^{(1)}(x) = 0$ for $x \in V$. Thus $b_1^{(1)}$ is given by $b_1^{(1)}(x) = b \cdot x$ for some $b \in V^\perp$ and (6) becomes for $q = 1$,

$$(17) \quad \text{spt } \nu \subset \{x : b \cdot x = |P_{V^\perp} x|^2\}.$$

Using some delicate algebraic calculations one can then prove that $b = 0$, see Preiss [4, 3.15]. This shows by (17) that ν is m -flat provided it is flat at infinity and, in particular, that all 2-uniform measures are 2-flat.

At this point there is essentially enough information to complete the proof of Theorem 17.8 also for $m \geq 3$. That is, we have now that the quantitative condition (16) implies that ν is m -flat. Thus we have a statement of the type “every m -uniform measure is either flat or far away from any flat measure” (this can also be formulated in terms of the metrics F_r defined in 14.12; see Preiss [4, 3.14]). Hence a connectedness argument applies as mentioned before.

Rectifiability and packing measures

We can now use Theorem 17.6 to continue from Theorem 6.12 to deduce much more about the sets for which Hausdorff and packing measures agree; this result was proven by Saint Raymond and Tricot [1].

17.11. Theorem. *Let $0 < s < n$ and $E \subset \mathbf{R}^n$ with $0 < \mathcal{P}^s(E) < \infty$. Then $\mathcal{P}^s(E) = \mathcal{H}^s(E)$ if and only if s is an integer and $\mathcal{P}^s \llcorner E$ is s -rectifiable.*

Proof. By the Borel regularity of $\mathcal{P}^s$ and $\mathcal{H}^s$ we may assume E to be a Borel set. If $\mathcal{P}^s(E) = \mathcal{H}^s(E)$, Theorem 6.12 says that $\Theta^s(E, x) = 1$ for $\mathcal{P}^s$, and by Theorem 5.12 also for $\mathcal{H}^s$, almost all $x \in E$. Hence s is an integer by Theorem 14.10 and E is s -rectifiable by Theorem 17.6. Since the equality $\mathcal{H}^s(E) = \mathcal{P}^s(E) < \infty$ and the inequality $\mathcal{H}^s \leq \mathcal{P}^s$ imply $\mathcal{P}^s(A) = 0$ for all $A \subset E$ with $\mathcal{H}^s(A) = 0$, $\mathcal{P}^s \llcorner E$ is also s -rectifiable.

Conversely, if $\mathcal{P}^s \llcorner E$ is s -rectifiable, then $\mathcal{P}^s \llcorner E \ll \mathcal{H}^s$ and we can use Theorem 17.6 similarly to conclude that $\Theta^s(E, x) = 1$ for $\mathcal{P}^s$ almost all $x \in E$. Hence $\mathcal{P}^s(E) = \mathcal{H}^s(E)$ by Theorem 6.12. $\square$

Remarks

17.12. (1) Let $h: (0, \infty) \rightarrow (0, \infty)$ be a non-decreasing function with $\lim_{r \downarrow 0} h(r) = 0$. Starting from the case $h(r) = r^s$ which was considered

above, one can ask more generally: for which functions h does there exist a non-zero Radon measure μ on $\mathbf{R}^n$ such that the limit

$$(1) \quad \lim_{r \downarrow 0} \frac{\mu(B(x, r))}{h(r)}$$

exists and is positive and finite for μ almost all $x \in \mathbf{R}^n$.

It was shown by Mattila [3] that in $\mathbf{R}^1$ this can happen only if the limit $\lim_{r \downarrow 0} (h(r)/r)$ exists and is positive and finite. In this case one can take for μ the measures which are absolutely continuous with respect to the Lebesgue measure, and only those. Preiss [4, §6] gave a great deal of information on such functions h in general dimensions showing in particular that near zero h must behave, in a weak sense, like r^m for some integer m . However, the precise statement generalizing the above one-dimensional result is false in $\mathbf{R}^n$ for $n \geq 2$. For example, for $h(r) = r/|\log r|$ there is a non-zero Radon measure μ on $\mathbf{R}^2$ for which the positive and finite h -density exists μ almost everywhere while for $h(r) = r|\log r|$ there is no such measure.

(2) N. A. Watson [1] applied Marstrand's and Preiss's theorems 14.10 and 17.8 to the behaviour of Gauss–Weierstrass and Poisson integrals. They are solutions of the heat and Laplace equations, respectively.

(3) David and Semmes derived the following density characterization of uniformly m -rectifiable sets for $m = 1, 2$, and $n - 1$, recall 15.23 and see David and Semmes [2, Theorems I.2.52 and I.2.56].

Let $m = 1, 2$, or $n - 1$ and let E be as in 15.23(1). For positive numbers C and ε denote by $\mathcal{G}_m(C, \varepsilon)$ the set of $(x, t) \in E \times (0, \infty)$ for which there exists a Borel measure μ such that $\text{spt } \mu = E$,

$$C^{-1}r^m \leq \mu(B(y, r)) \leq Cr^m \quad \text{for } y \in E \text{ and } r > 0,$$

and

$$|\mu(B(y, s)) - s^m| \leq \varepsilon t^m$$

for all $y \in E \cap B(x, t)$ and $0 < s < t$. Then E is uniformly m -rectifiable if and only if there exists C such that $(E \times (0, \infty)) \setminus \mathcal{G}_m(C, \varepsilon)$ is a Carleson set (recall 15.23 (3)) for every $\varepsilon > 0$.

The proof of the “if” part uses the results of Preiss, and Kowalski and Preiss, described before. For $2 < m < n - 1$ no such characterization is known. This is due to the fact that there is not enough information about m -uniform measures in this case. The “only if” part, and somewhat more, holds for all m .

Exercises.

1. Let ν be an m -uniform measure such that $\text{spt } \nu$ is m -rectifiable and $\nu(B(x, r)) = (2r)^m$ for $x \in \text{spt } \nu$, $0 < r < \infty$. Show that $\nu = \mathcal{H}^m \llcorner \text{spt } \nu$. Hint: Use Theorems 16.2(1) and 2.12.
2. Show by a direct computation that if Γ is a smooth curve such that $\mathcal{H}^1 \llcorner \Gamma$ is 1-uniform, then Γ is a line.
3. Let μ be a Radon measure on $\mathbf{R}$. Show that at μ almost all points $a \in \mathbf{R}$,

$$\text{Tan}(\mu, a) \neq \{c\delta_0 : 0 < c < \infty\} \cup \{c\mathcal{L}^1 : 0 < c < \infty\}.$$

Hint: Pick small r_i such that $c_i T_{a, r_i \#} \mu$ looks like δ_0 , then the “first” $r > r_i$ such that $cT_{a, r \#} \mu$ is not close to δ_0 in some suitable quantitative sense, and show that $cT_{a, r \#} \mu$ cannot yet be close to $\mathcal{L}^1$.

4. Let μ be a Radon measure on $\mathbf{R}^n$ and E the set of those $x \in \mathbf{R}^n$ for which $\Theta^m(\mu, x)$ exists and is positive and finite. Show that if $\mu(\mathbf{R}^n \setminus E) = 0$, then $\mu(B) = \int_{B \cap E} \Theta^m(\mu, x) d\mathcal{H}^m x$ for Borel sets $B \subset \mathbf{R}^n$. Hint: You may use the differentiation theory of Chapter 2.
5. Let ν be a 2-uniform measure on $\mathbf{R}^3$ such that $\text{spt } \nu$ is contained in a finite union of planes and spheres. Show that ν is 2-flat.